CLASS XI CHEMISTRY

Some Basic Concepts of Chemistry

Notation note: Chemical formulas use proper subscripts throughout. Scientific powers are shown in readable plain-text form where needed.

1. Atoms and molecules

Atom

An atom is the smallest unit of an element that takes part in a chemical reaction. Atoms contain protons (positive charge),

neutrons (no charge) and electrons (negative charge). Protons and neutrons are in the nucleus; electrons occupy the region

around it. For example, a copper wire contains a huge number of copper atoms.

Molecule

A molecule is a group of two or more atoms chemically bonded together that can exist as a separate unit. H_2 contains two

hydrogen atoms; O_2 contains two oxygen atoms; H_2O contains two hydrogen atoms and one oxygen atom; CO_2 contains one

carbon atom and two oxygen atoms. In a formula, a number written after an element symbol tells how many atoms of that

element are present. If no number is written, it means one.

Ionic compounds

Not every substance consists of separate molecules. Solid sodium chloride (NaCl) forms a repeating arrangement of sodium and

chloride ions. NaCl shows the simplest ratio of ions, so it is described as a formula unit rather than a separate molecule.

2. Nature and classification of matter

Matter

Matter is anything that has mass and occupies space. Examples include a book, water, air and a stone. Matter is made of tiny

particles such as atoms, molecules or ions.

States of matter

Solid: particles are close and mainly vibrate around fixed positions, so the solid has a fixed shape and volume. Liquid: particles

remain close but move past each other, so a liquid has a fixed volume but takes its container's shape. Gas: particles are far apart

and move rapidly, so a gas has no fixed shape or volume and fills its container.

Changes of state

Melting: solid to liquid, as ice becomes water. Freezing: liquid to solid. Vaporisation: liquid to gas. Condensation: gas to liquid, as

vapour forms droplets. Sublimation: solid to gas directly, as dry ice can do under suitable conditions.

Pure substances and mixtures

A pure substance has a fixed chemical composition. An element contains one kind of atom, e.g. Fe or Cu. A compound contains

different elements chemically combined in a fixed ratio, e.g. H_2O or CO_2. A mixture contains substances physically mixed in

variable proportions. A homogeneous mixture is uniform throughout, e.g. salt solution or air. A heterogeneous mixture is not

uniform, e.g. oil and water.

3. Properties of matter and measurement

Physical properties

Physical properties can be observed or measured without changing the chemical identity of a substance. Examples include

colour, density, melting point and boiling point.

Chemical properties

Chemical properties describe how a substance reacts to form new substances. For example, iron reacts with oxygen and

moisture to form rust; wood burns and forms new substances.

Measurement

A measurement includes a numerical value and a unit. For example, 5 cm means five centimetres. Units make measurements

clear and comparable.

4. SI base units, derived units and prefixes

Seven SI base units

Length: metre (m); mass: kilogram (kg); time: second (s); electric current: ampere (A); thermodynamic temperature: kelvin (K);

amount of substance: mole (mol); luminous intensity: candela (cd). The SI base unit of mass is kilogram, although grams are

common in chemistry.

Derived units

Derived units combine base units. Area uses m^2; volume uses m^3; speed uses m s^(-1); density uses kg m^(-3); force uses

newton (N), equal to kg m s^(-2). Chemistry also commonly uses litres (L), millilitres (mL) and grams (g).

Prefixes

kilo (k) = 10^3; centi (c) = 10^(-2); milli (m) = 10^(-3); micro (u) = 10^(-6); nano (n) = 10^(-9). Thus, 1 km = 1000 m, 1 cm = 0.01

m, 1 mm = 0.001 m, and 1 L = 1000 mL.

5. Mass, weight, volume, density and temperature

Mass and weight

Mass measures the amount of matter. Its SI unit is kilogram (kg), though grams (g) are often used in chemistry. Mass does not

change when an object moves from Earth to the Moon. Weight is gravitational force: W = m x g, where m is mass and g is

gravitational acceleration. Weight is measured in newtons (N) and changes with gravity.

Volume

Volume is the space occupied by an object or substance. Its SI unit is m^3. Useful conversions: 1 L = 1000 mL; 1 mL = 1 cm^3; 1

L = 1000 cm^3. For a box, volume = length x breadth x height. For an irregular solid that does not dissolve or react in water, the

increase in water level after immersion equals its volume.

Density

Density is mass per unit volume: density = mass / volume. Common units are g cm^(-3), g mL^(-1) and kg m^(-3). Example: mass

= 60 g and volume = 20 cm^3. Density = 60 / 20 = 3 g cm^(-3). Rearranging gives mass = density x volume and volume = mass /

density.

Temperature

Temperature indicates how hot or cold something is and is related to the average kinetic energy of its particles. SI unit: kelvin (K).

T(K) = t(degrees C) + 273.15; t(degrees C) = T(K) - 273.15. Example: 27 degrees C = 300.15 K, approximately 300 K.

6. Uncertainty and scientific notation

Measurement uncertainty

No measurement is perfectly exact. Instruments have limited precision. A ruler marked only in centimetres cannot measure a

length as finely as a ruler marked in millimetres.

Scientific notation

Write a number as a x 10^n, where a is at least 1 but less than 10. Large number: 450000 = 4.5 x 10^5. Small number: 0.00072 =

7.2 x 10^(-4). The exponent tells how many places the decimal point moves.

Multiplication

Multiply the coefficients and add exponents: (2 x 10^3) x (3 x 10^4) = 6 x 10^(3+4) = 6 x 10^7.

Division

Divide the coefficients and subtract exponents: (8 x 10^7) / (2 x 10^3) = 4 x 10^(7-3) = 4 x 10^4.

Addition and subtraction

First make the powers of 10 the same. Example: 3 x 10^5 + 2 x 10^4 = 3 x 10^5 + 0.2 x 10^5 = 3.2 x 10^5. Do not add or subtract

exponents when adding or subtracting numbers.

7. Significant figures

Rules

All non-zero digits count: 347 has 3 significant figures. Zeros between non-zero digits count: 1005 has 4. Leading zeros do not

count: 0.0042 has 2. Trailing zeros after a decimal count: 2.500 has 4. Trailing zeros in whole numbers can be ambiguous, so

scientific notation can clarify precision: 1.5 x 10^3 has 2 significant figures; 1.500 x 10^3 has 4.

Multiplication and division

Round the final answer to the same number of significant figures as the input with the fewest. Example: 2.5 x 3.42 = 8.55, which

rounds to 8.6 (2 significant figures).

Addition and subtraction

Round to the least precise decimal place. Example: 12.11 + 3.2 = 15.31, which rounds to 15.3. Keep extra digits in intermediate

steps and round at the end.

8. Dimensional analysis

Unit conversion

Use a conversion fraction equal to 1, arranged so unwanted units cancel. Example: 2.5 km x (1000 m / 1 km) = 2500 m. Example:

500 mL x (1 L / 1000 mL) = 0.5 L. Check that the final unit matches the requested quantity.

9. Laws of chemical combinations

Conservation of mass

Mass is neither created nor destroyed during an ordinary chemical reaction. In a closed system, total mass of reactants equals

total mass of products. Example: 12 g carbon + 32 g oxygen gives 44 g carbon dioxide.

Definite proportions

A pure compound always contains the same elements in the same fixed proportion by mass. Water has hydrogen and oxygen in

a mass ratio of 1:8; 9 g water contains 1 g hydrogen and 8 g oxygen.

Multiple proportions

When two elements form more than one compound, masses of one element that combine with a fixed mass of the other are in a

simple whole-number ratio. For 12 g carbon, CO contains 16 g oxygen and CO_2 contains 32 g oxygen. Ratio = 1:2.

Gay-Lussac's law

Reacting gases and gaseous products have simple whole-number volume ratios at the same temperature and pressure. $2H_2(g)$

+ $O_2(g)$ -> $2H_2O(g)$ has volume ratio 2:1:2 under suitable conditions.

Avogadro's law

Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules. At constant temperature

and pressure, gas volume is proportional to moles: V is proportional to n.

10. Dalton's atomic theory

Main ideas

Matter is made of atoms. Atoms of different elements have different properties. Compounds form when atoms combine in simple

whole-number ratios. Ordinary chemical reactions rearrange atoms rather than creating or destroying them.

Modern correction

Atoms of the same element can have different masses because of isotopes. Atoms contain protons, neutrons and electrons, so

some original statements of Dalton's theory have been revised.

11. Atomic, average atomic, molecular and formula masses

Unified atomic mass unit (u or amu)

The unified atomic mass unit, symbol u, is exactly one-twelfth of the mass of one carbon-12 atom. It is also commonly called the

atomic mass unit (amu). Its approximate value is 1.6605 x 10^(-27) kg. This unit is convenient because atoms are extremely

small. Atomic and molecular masses are commonly expressed in u.

Atomic mass

Atomic mass is the mass of an atom expressed in u. Approximate values: H = 1 u, C = 12 u, N = 14 u, O = 16 u, Na = 23 u and Cl

= 35.5 u.

Isotopes and average atomic mass

Isotopes have the same number of protons but different numbers of neutrons. Average atomic mass is a weighted average: add

each isotope's mass multiplied by its fractional abundance. Example: 10 u at 20% and 11 u at 80% gives 10 x 0.20 + 11 x 0.80 =

10.8 u.

Molecular mass

Add the atomic masses of all atoms in one molecule. H_2O = 2 x 1 + 16 = 18 u. CO_2 = 12 + 2 x 16 = 44 u.

Formula mass

Add the atomic masses in a formula unit, especially for ionic compounds. NaCl = 23 + 35.5 = 58.5 u.

Molar mass difference

Atomic or molecular mass is the mass of one atom or molecule in u. Molar mass is the mass of one mole, usually in g mol^(-1). A

water molecule has mass about 18 u, while one mole of water has molar mass about 18 g mol^(-1).

12. Mole concept and molar mass

Why use the mole?

Atoms and molecules are too small to count individually in ordinary laboratory work. A mole is a counting unit, just as a dozen

means 12 items.

Avogadro constant

One mole contains exactly 6.02214076 x 10^23 specified entities. School calculations commonly use 6.022 x 10^23. This is the

Avogadro constant, N_A. The entities must be specified: atoms, molecules, ions, etc. One mole of O_2 contains 6.022 x 10^23 O_2

molecules, equal to 2 moles of oxygen atoms.

Molar mass and formulas

Molar mass is the mass of one mole, usually in g mol^(-1). Moles = mass / molar mass. Number of particles = moles x N_A.

Moles = number of particles / N_A.

Example: moles from mass

Find moles in 36 g water. Molar mass H_2O = 18 g mol^(-1). Moles = 36 / 18 = 2 mol.

Example: particles from moles

Number of molecules in 2 mol water = 2 x 6.022 x 10^23 = 1.2044 x 10^24 molecules.

Example: mass from moles

Mass of 0.5 mol CO_2 = 0.5 x 44 = 22 g.

13. Percentage composition

Formula

Percentage by mass of an element = (mass of that element in one mole of compound / molar mass of compound) x 100.

Example: water

Molar mass H_2O = 18 g mol^(-1). Hydrogen percentage = (2/18) x 100 = 11.11%. Oxygen percentage = (16/18) x 100 = 88.89%.

Total = 100%.

14. Empirical and molecular formulae

Empirical formula

The empirical formula gives the simplest whole-number ratio of atoms. $C_6H_{12}O_6$ simplifies to CH_2O; H_2O_2 simplifies to HO.

Finding empirical formula

Assume 100 g if percentages are given, so percentages become grams. Divide each mass by its atomic mass to find moles.

Divide all mole values by the smallest. Convert the ratios to whole numbers; multiply all ratios by a small integer if necessary.

Example

A compound has 40% C, 6.67% H and 53.33% O. In 100 g: C = 40/12 = about 3.33 mol; H = 6.67/1 = 6.67 mol; O = 53.33/16 =

about 3.33 mol. Divide by 3.33 to get 1:2:1. Empirical formula = CH_2O.

Molecular formula

Molecular formula gives the actual number of atoms in a molecule. Multiplier n = molecular molar mass / empirical formula mass.

Multiply each empirical-formula subscript by n. Example: CH_2O has formula mass 30. If molecular molar mass is 180, n = 180/30

= 6, so molecular formula = $C_6H_{12}O_6$.

15. Stoichiometry and limiting reagent

Stoichiometry

Stoichiometry calculates the amounts of reactants and products using a balanced chemical equation.

Balanced equations

A balanced equation has equal numbers of each element's atoms on both sides. $2H_2 + O_2$ -> $2H_2O$ has 4 H atoms and 2 O

atoms on each side. Coefficients give the mole ratio 2:1:2. Change coefficients, never subscripts, to balance equations.

Calculation steps

1. Balance the equation. 2. Convert the known mass to moles. 3. Use the coefficient ratio. 4. Convert required moles to mass if

needed. 5. Include units.

Example

4 g H_2 = 4/2 = 2 mol H_2. With excess oxygen, the H_2:H_2O ratio is 1:1, so 2 mol water forms. Mass = 2 x 18 = 36 g, assuming

complete reaction.

Limiting reagent

The limiting reagent is used up first and limits the amount of product. The excess reagent remains. For $2H_2 + O_2$ -> $2H_2O$, 4 mol

H_2 and 1 mol O_2 make 2 mol H_2O; O_2 is limiting and 2 mol H_2 remain.

16. Concentration of solutions

Solution, solute and solvent

A solution is a homogeneous mixture. The solute is the substance dissolved; the solvent does the dissolving. In salt water, salt is

solute and water is solvent.

Mass percent

Mass % = (mass of solute / mass of solution) x 100. Mass of solution = mass of solute + mass of solvent. Example: 10 g salt + 90

g water gives 100 g solution, so salt mass percent = 10%.

Mole fraction

Mole fraction of A = moles of A / total moles. For A and B: X_A = n_A/(n_A+n_B); X_B = n_B/(n_A+n_B). All mole fractions add

to 1. Example: 2 mol A and 3 mol B gives X_A = 0.4 and X_B = 0.6. Mole fraction has no unit.

Molarity

Molarity = moles of solute / volume of solution in litres. Unit: mol L^(-1), also written M. Example: 0.5 mol in 250 mL solution; 250

mL = 0.250 L, so molarity = 0.5/0.250 = 2 mol L^(-1). Use solution volume, not solvent volume. Molarity can change with

temperature.

Molality

Molality = moles of solute / mass of solvent in kg. Unit: mol kg^(-1). Example: 0.5 mol solute in 250 g = 0.250 kg solvent gives

molality = 0.5/0.250 = 2 mol kg^(-1). Use solvent mass, not solution mass. Molality does not change with temperature.

Molarity versus molality

Molarity uses volume of solution and may change with temperature. Molality uses mass of solvent and does not change with

temperature.

17. Quick formula sheet

Core formulas

Density = mass / volume. Weight = mass x gravitational acceleration. T(K) = t(degrees C) + 273.15. Moles = mass / molar mass.

Particles = moles x Avogadro constant. Percentage composition = (mass of element / mass of compound) x 100.

Molecular-formula multiplier = molecular molar mass / empirical formula mass. Mass % = (mass of solute / mass of solution) x

100. Mole fraction = moles of component / total moles. Molarity = moles of solute / litres of solution. Molality = moles of solute / kg

of solvent.

Common distinctions

Mass vs weight: amount of matter vs gravitational force. Atomic/molecular mass uses u; molar mass uses g mol^(-1). Empirical

formula gives the simplest ratio; molecular formula gives actual atom counts. Limiting reagent runs out first. Molarity uses solution

volume; molality uses solvent mass. Multiplication/division use significant figures; addition/subtraction use decimal-place

precision.